

PCBSMITH

Advanced PCB Design Research and Source Guide

A context-gated rule framework for power integrity, high-speed interfaces, DDR, RF, mixed signal, thermal, isolation, advanced packaging, manufacturability, and component evidence.

PINNED SOURCES

13

Official PDFs validated and kept in the private cache.

TYPED BLOCKS

5

Official endpoints retained as explicit network failures.

RULE FAMILIES

23

Research catalogue; no blanket production promotion.

Core conclusion: advanced PCB design cannot be reduced to a universal checklist of distances. Rules must be conditioned on the exact part, interface, topology, stack-up, load, environment, manufacturing process, and validation method.

Prepared 2026-07-22

Research complete; production-rule promotion and advanced simulation remain open.

Executive assessment

The missing knowledge is not one hidden master checklist. Professional PCB work combines several different authorities:

1. The exact component datasheet, errata, hardware-design guide, and reference design.
2. The selected interface, product-safety, fabrication, and assembly standards.
3. The selected fabricator stack-up and assembler process.
4. Calculations and simulations using project-specific electrical and mechanical inputs.
5. Post-layout review, measurement, and first-article evidence.

Books and generic application notes teach mechanisms and reusable review questions. They do not safely supply universal dimensions for every design. Community reviews and videos are useful discovery channels, especially for failure modes, but a claim found there must be traced to a primary source or kept as an advisory hypothesis.

The recommended direction for PCBSmith is therefore a context-gated rule system, not a larger flat list of magic numbers. Each promoted rule must state:

- its exact applicability predicate;
- required project inputs;
- authority and revision;
- whether it is hard, derived, advisory, or simulation-bound;
- the implementation consumer;
- the verification method;
- confidence and unresolved conflicts.

What was found locally

The local collection is already unusually strong for foundational PCB work. It contains IPC-7352, IPC-7093A, IPC-2152, IPC-2221A, IPC-2222, USB 2.0 and USB Type-C material, Sensirion design-in guidance, ESD and thermal sources, EMC and signal-integrity books, and substantial existing distillation. The main problem is utilization: many sources are pinned or summarized but not yet reconciled with exact parts, converted into typed rules, bound to consumers, firing-tested, or production-exercised.

The current code also already has partial semantic structures for decoupling loops, switching hot loops, thermal intent, antenna constraints, oscillator zones, sensor isolation, and return adjacency. Those are important foundations, but they do not yet form a complete advanced-board workflow.

Acquisition result

The approved intake pass selected 18 free official documents across MCU hardware, PDN, mixed-signal, high-speed, DDR, isolation, motor drive, RF, hot-swap, switching power, oscillator design, and Ethernet.

- 13 documents were downloaded through the repository's allowlisted intake, validated as PDFs, SHA-256 pinned, and stored only in the private cache.

- 5 Analog Devices URLs remained typed `network_error` results after bounded

retries. Their official pages remain monitored sources, but the payloads were not silently replaced by third-party mirrors.

- TI SLVA680A was already local and was not downloaded again.

- The complete public identity projection is in

data/advanced-pcb-source-intake-2026-07-22.json.

This is evidence that automatic source retrieval exists but is not yet robust enough for unattended production use. It needs retry with jitter, redirect and content-disposition handling, browser-capable official-host fallback, rate-limit awareness, and a resumable queue. Typed blocked states must remain visible.

Rule authority and promotion model

Authority order

Use the following default precedence, while still checking for project-specific exceptions:

1. Applicable product safety or regulatory standard.
2. Selected fabricator and assembler capability profile.
3. Exact component datasheet, errata, and manufacturer design guide.
4. Exact manufacturer reference design with matching part, package, and

topology.

5. Interface or device-family guide.
6. Industry standard.
7. Generic application note or textbook.
8. Community review, forum, or video.

Promotion ladder

found -> identity verified -> revision pinned -> relevant section distilled -> conflicts reconciled -> typed rule proposed -> implementation bound -> firing-tested -> production-exercised

Downloading or summarizing stops near the beginning of this ladder. It is not proof that a board generator uses the information.

Four rule classes

- Hard: fail-closed once its declared applicability context is present.
- Derived: a numerical constraint computed from project inputs such as stack-up, timing, current, load step, or altitude.
- Advisory: a review prompt that needs judgment or measurement.
- Simulation-bound: cannot be accepted from geometry alone; requires a named model, solver, correlation, or laboratory method.

Expanded design-rule catalogue

The machine-readable catalogue defines 23 rule families. The most important findings are summarized below.

1. Architecture and exact-part identity

- Resolve every power, ground, exposed pad, reset, boot, programming, reference, and unused pin from exact manufacturer evidence.
- Treat package migration and alternate parts as new evidence events. Similar names do not imply compatible pin, pad, thermal, or layout requirements.
- Record every reference-design divergence and its likely consequence.
- Create a bring-up and recovery plan while the schematic is still editable.

2. Power tree, current budget, sequencing, and reset

- Budget static and transient current by operating mode, not only average current.
- Declare rail tolerances, startup order, ramp limits, brownout behavior, discharge, and power-good dependencies.
- Check that unpowered domains are not back-powered through I/O, protection diodes, pull-ups, or level shifters.
- Re-run the power tree after layout and component selection; plane necks and regulator thermal limits can invalidate an earlier schematic budget.

3. Power entry, inrush, surge, and fault energy

- Treat connector rating, cable/source impedance, input capacitance, fuse, reverse polarity, TVS, hot-swap element, and downstream absolute maximum as one protection chain.
- Check MOSFET transient safe operating area and pulse energy. A DC current rating is not an inrush proof.
- Model where regenerative or fault energy goes. This is essential for motors, inductive loads, long cables, and hot-plug systems.

4. PDN and decoupling

- Minimize the complete capacitor-current loop. Straight-line capacitor-to-pin distance is only a proxy and can be misleading across vias and planes.
- Derive target impedance from allowed droop divided by load step over a declared frequency/time range. Do not use one target for every rail.
- Use capacitor MPN behavior: DC-bias capacitance loss, ESR, ESL, package, mounting inductance, and tolerance matter.
- Review anti-resonance, plane resonance, VRM control response, package inductance, and power-via current density for complex ICs.

5. Stack-up, impedance, and return structure

- Obtain a fabricator-approved stack-up before freezing controlled-impedance geometry.
- Recalculate width and spacing when dielectric, copper, plating, solder mask, or impedance tolerance changes. Values copied from a vendor reference stack are not portable.
- Declare the reference plane for each critical route and provide a nearby return transition when the signal changes reference.
- Consider copper roughness, D_k/D_f , glass weave, and loss budget when interface speed and reach make them material.

6. Return current, grounding, and zoning

- Partition by current path, susceptibility, and energy - not by arbitrary analog/digital labels.
- Do not route critical signals over plane gaps. A split ground can become a slot antenna and force a large return loop.
- If a layer transition changes the reference potential, design the return transition intentionally; a distant decoupling capacitor is not equivalent to a colocated low-inductance path.
- Keep chassis/shield current paths separate from sensitive logic where the interface and product architecture require it.

7. Clocks, crystals, PLLs, and jitter

- Select the exact crystal/oscillator and calculate load capacitance including pin and board parasitics.
- Check drive level, startup time, negative-resistance or transconductance margin, temperature, supply variation, and probe loading.
- Treat SerDes and converter reference clocks as noise-sensitive analog paths with explicit phase-noise/jitter and supply-noise budgets.
- Do not invent a universal ground split around a crystal. Follow the exact IC and oscillator guidance and preserve a controlled quiet return.

8. High-speed digital and SerDes

- Edge rate, not only clock frequency, determines whether a route behaves as a transmission line.
- Controlled impedance, continuous reference, pair symmetry, equal transitions, connector/package behavior, and protocol-specific loss/skew limits take precedence over aesthetic routing.
- Do not maximize differential-pair coupling blindly. Follow the interface and stack-up-specific impedance/crosstalk solution.
- Include package delay before board-level length matching.
- At multi-gigabit rates, review via stubs, backdrilling, antipads, AC coupling, fiber weave, insertion loss, return loss, crosstalk, and mode conversion.

- Critical-channel test points can be harmful stubs; use planned observation methods instead of universal probe pads.

9. DDR and source-synchronous memory

- There is no safe universal DDR length-match number. Rules depend on the processor, memory, topology, rank count, package delays, ODT/drive settings, stack-up, and training behavior.
- Preserve vendor-defined byte lanes, timing groups, topology, placement, termination, and keepouts.
- Run pre-layout topology analysis and post-layout timing/SI simulation for deviations or advanced data rates.
- Capture training margins during bring-up and compare them with the design assumptions.

10. Ethernet PHY and transformer interfaces

- Bind the PHY, magnetics, connector shield, center taps, terminations, chassis strategy, and ESD path as one interface block.
- Follow the exact PHY checklist for PHY-to-magnetics placement, MDI routing, pair separation, clocking, and power.
- Inspect metal keepouts around magnetics and the connector, as well as return vias and shield-current paths.

11. RF and antennas

- Prefer the exact manufacturer reference layout and record deviations.
- Place antennas with the required board-edge relationship and all-layer keepout. Include the enclosure, battery, display, cables, metal, and nearby human body in the mechanical/RF review.
- Derive the feed geometry from the selected stack-up and provide an accessible matching network.
- Final tuning must use the assembled enclosure. A bare-board 50-ohm trace does not prove radiated performance.

12. Switching power

- Identify topology-specific high di/dt hot loops before placement and minimize their actual copper loop area.
- Bound switch-node copper. Larger copper may help heat spreading while making capacitive coupling and EMI worse; the tradeoff must be explicit.
- Use Kelvin feedback and current sense. Keep their reference away from load copper voltage drop and switching fields.
- Control gate-source current loops and consider footprints for damping or snubbing where evidence supports them.

13. Motors and high-current switching

- Calculate stall, startup, regenerative, and fault cases, not only nominal running current.
- Separate shunt force and sense connections and check amplifier common-mode transients.
- Review phase symmetry, gate loops, dead time, shoot-through, connector/cable ratings, thermal spreading, and via arrays.
- Use real copper thickness, plating, temperature rise, voltage drop, and duty cycle. Avoid a fixed amps-per-via rule.

14. Precision analog and mixed signal

- Build a noise and error budget before turning generic layout advice into constraints.
- Control converter sampling-current paths, reference decoupling, high-impedance leakage, input protection capacitance, thermal gradients, and digital return currents.
- Consider guards, shields, matched thermal placement, and contamination rules where the source impedance or accuracy justifies them.
- A single continuous ground plane is often safer than an arbitrary split; separation should come from placement and current-path control.

15. Sensors and environmental coupling

- Model how the measurand reaches the sensor: airflow, thermal path, pressure port, sound aperture, magnetic field, board motion, or strain.
- Check self-heating, duty cycle, nearby heat sources, copper thermal bridges, contamination, coating, enclosure openings, and mechanical stress.
- Slots and copper removal are tools, not automatic improvements. They need response-time, strength, EMC, and manufacturing review.

16. Thermal design and current density

- Treat θ_{JA} as board- and environment-dependent. Do not apply a package table value as a universal temperature prediction.
- Evaluate steady and transient loss, ambient, airflow, copper, vias, enclosure, contact paths, component derating, and safe operating area.
- Check both temperature rise and voltage drop. IPC-2152 is useful local historical evidence, but it should not become one universal equation.
- Coordinate thermal-via geometry with paste, voiding, assembly, and BGA/BTC inspection constraints.

17. ESD, EFT, surge, and EMC

- Arrange external connector, protection element, filtering, and protected IC

in current-flow order.

- Minimize unprotected trace length and the inductance of the discharge return.
- Keep the ESD path from coupling into protected circuitry or forcing shield

current through logic ground.

- Check protection capacitance and clamping behavior against interface signal quality and the actual threat current.

18. Isolation and safety

- Derive creepage and clearance from the applicable product standard, working and transient voltage, insulation class, pollution degree, material group, altitude, and process.
- Check the barrier on every copper and conductive mechanical layer.
- Credit slots or conformal coating only where the selected standard and controlled manufacturing process allow it.
- Coordinate isolation with EMI; a capacitive return intended to reduce emissions must preserve safety requirements.

19. BGA, HDI, microvias, and via-in-pad

- Prove fanout and escape feasibility before locking the package and layer count.
- Bind via technology, lamination cycles, fill/cap process, capture lands, reliability class, assembly, X-ray, and rework into the board process profile.
- Do not use unfilled via-in-pad under soldered pads unless the exact assembly process explicitly supports it.
- Coordinate BGA breakout with return vias, PDN access, decoupling, and channel discontinuity rather than solving escape in isolation.

20. Fabrication, assembly, and finish

- Replace generic 6/6 mil assumptions with selected fabricator and assembler profiles.
- Treat solder mask, paste apertures, stencil thickness, panelization, fiducials, breakaways, component overhang, and depanel stress as first-class design inputs.
- Select surface finish for pitch, contact use, storage, assembly, and reliability. ENIG is not automatically best for every board.
- Run both fabricator and assembler DFM before release.

21. Test, debug, programming, and bring-up

- Preserve programming, recovery, reset, and boot visibility.
- Add measurement access for power rails and low-speed controls, but do not add stubs to critical channels without a channel budget.
- Consider boundary scan and built-in self-test for BGA-heavy designs.

- Define current-limited first power, rail order, expected readings, and stop conditions before hardware arrives.

22. Reliability, environment, and lifecycle

- Use a mission profile for temperature, humidity, shock, vibration, duty, service life, and allowed failure rate.
- Apply voltage, current, power, temperature, and transient derating by component type and environment.
- Include MLCC DC-bias loss and flex cracking, contamination and CAF risk, coating compatibility, component lifecycle, PCNs, and replacement strategy.

23. CAD assets and 3D models

- Use the cascade: exact manufacturer assets, official KiCad libraries, licensed CAD provider, then verified custom asset.
- Reconcile symbol pins with the datasheet and footprint geometry with the manufacturer package drawing. A model downloaded by part number is not proof of correctness.
- Align STEP models 1:1 in millimeters and label fidelity as exact package, complete module, connector-only, or proxy.
- Visually inspect every imported symbol, footprint, and 3D model before it can become an authority.

Complexity escalation for future boards

L0 - simple low-speed two-layer board

Required: exact-part evidence, support-circuit audit, ERC/DRC, selected fabrication profile, standard visual review, and a basic bring-up plan.

L1 - switching, mixed-signal, RF module, motor, or external interfaces

Add: current-path and return-path declarations, topology placement zones, thermal/transient calculations, focused layer-isolated renders, and a bench validation plan.

L2 - controlled impedance, BGA, DDR, isolation, or multi-rail SoC

Add: fabricator stack-up signoff, package escape feasibility, PDN/SI simulation plan, sequencing, formal bring-up, and assembler DFM.

L3 - multi-gigabit, FPGA/RFSoc, HDI, safety-critical, or certification-bound

Add: channel and PDN simulation with controlled models, a product-standard compliance matrix, thermal/airflow simulation, laboratory pre-compliance, and a formal unresolved-risk register. PCBSmith should generate and audit artifacts at this level, but it must not claim autonomous engineering signoff.

Component and document retrieval architecture

What an API can and cannot provide

Distributor and aggregation APIs can help find:

- exact manufacturer identity and orderable MPN;
- lifecycle and supply information;
- manufacturer datasheet URL;
- product image and sometimes CAD-provider links;
- parametric data for preliminary screening.

They generally do not provide trustworthy, context-specific PCB layout rules. Those come from the manufacturer datasheet, hardware guide, errata, reference design, package drawing, and interface documents.

Recommended cascade

1. Normalize manufacturer and exact MPN.
2. Search the local evidence and asset cache by identity and revision.
3. Query the manufacturer source or approved official URL.
4. Use a distributor API only as a metadata and official-datasheet locator.
5. Search official KiCad libraries for symbols, footprints, and 3D models.
6. Use a licensed CAD provider only after its terms permit the intended cache, transformation, and redistribution behavior.
7. Build a custom asset from the package drawing if no safe source exists.
8. Validate pins, dimensions, units, origin, rotation, side, height, and license.
9. Store private payload separately from commit-safe hashes and metadata.

Provider assessment

Provider	Useful data	Access	Assessment
DigiKey Product Information V4	MPN, manufacturer, product data, DatasheetUrl	OAuth and organization/account setup	Good metadata and datasheet locator; needs user credentials
Mouser Search API	MPN search, datasheet URL, lifecycle, image, availability, pricing	Account and API key; published call limits	Good fallback locator; not a layout-rule source
element14 Product Search API	Keyword, product, and MPN search; JSON/XML	API key	Useful additional locator
Nexar/Octopart	GraphQL part, supply, technical, and datasheet metadata	Account, application, OAuth, plan limits	Client exists locally but no credentials; terms restrict caching and self-hosting
KiCad official libraries	Symbols, footprints, and 3D models	Open GitLab repositories	Preferred generic CAD source; still requires exact-part verification
SnapMagic	Symbols, footprints, 3D assets	Commercial terms/API integrations	Conditional; licensing and cache rights must be approved first
Ultra Librarian	Symbols, footprints, 3D assets	Registration and terms	Conditional/manual unless a licensed integration is available

No provider should be called silently with user credentials, and no provider payload should be committed unless its license permits redistribution.

Remaining source gaps

High-value paid or conditional standards not found locally

These are future purchase or user-supplied candidates, not blockers for the current proof of concept:

- IPC-2226A - HDI and microvia design.
- IPC-7095D with Amendment 1 - BGA design and assembly.
- IPC-4761 - via protection structures.
- IPC-6012F - rigid-board qualification and performance.
- IPC-4552B - ENIG finish requirements.

The 2026-07-22 user intake added the original IPC-2226, IPC-6012C, and IPC-6013D. These materially improve historical and mechanism coverage, but do not close the current IPC-2226A, IPC-6012F, or IPC-6013E gaps. It also added an IPC-7251 first working draft for through-hole land-pattern research; its draft status must remain attached to every derived claim.

For advanced future directions also monitor the current revisions of IPC-2222, IPC-2223 for flex/rigid-flex, IPC-6013 for flex/rigid-flex qualification, IPC-6018 for high-frequency boards, IPC-A-610 and J-STD-001 for assembly acceptance/process, and applicable automotive, medical, aerospace, or functional-safety standards. Do not buy all of these preemptively; acquire them when a concrete project creates an applicability need.

Free current advanced references to monitor

- AMD UG583 for FPGA/SoC PDN, DDR, transceivers, RFSoc, and interface rules.
- Intel Agilex 5 PCB Design Guidelines for BGA escape, HSSI, EMIF, MIPI, PDN, fiber weave, vias, and backdrilling.
- The exact processor/FPGA hardware guide and memory-layout guide selected by a future project.
- The exact safety and EMC standards selected by the product category and market.

Proposed implementation sequence

1. Build a revision-aware source registry and resilient retrieval queue.
2. Add exact-part document-role discovery: datasheet, errata, hardware guide, package drawing, reference design, simulation model, CAD asset.
3. Introduce context schemas for stack-up, environment, interfaces, power tree, timing groups, protection threats, manufacturing process, and validation plan.
4. Promote only a small first set of rules with named consumers:
decoupling-loop topology, connector-to-ESD ordering, oscillator-zone evidence, switcher hot-loop membership, and stack-up/reference continuity.
5. Add project complexity classification L0-L3 and require progressively deeper gates.
6. Build simulation adapters only after their inputs and acceptance metrics are typed: impedance/field solver, IBIS/S-parameter SI, PDN, thermal, and power transient.
7. Expand the visual package with stack-up, return-path, power-domain, thermal, high-speed, and DFM diagnostic views.
8. Exercise each promoted rule on a materially different board and retain failure evidence before default adoption.

Important non-rules

The following statements must not become unconditional global checks:

- "Put every decoupling capacitor within X millimeters."
- "Always split analog and digital ground."
- "Route every differential pair as tightly coupled as possible."
- "Match all high-speed traces to the same length."
- "A via carries X amps."
- "Use one ounce copper and this trace-width equation for every board."
- "Theta-JA predicts this board temperature."
- "A 50-ohm trace guarantees an antenna will work."
- "Six mil design rules are safe for every fab."
- "A downloaded KiCad footprint or 3D model is correct because the MPN matches."

Each may be a useful heuristic under a declared context. None is a universal engineering authority.

Decision

This research should be added as a new sequential roadmap phase rather than retroactively inserted into completed phases. The research catalogue and source intake can be marked complete. Rule reconciliation, implementation, simulation, and production exercise remain open and should advance incrementally with future project needs.

Source and provider directory

The link text below opens the official source. Local state records whether the payload is pinned, monitored, conditional, or unavailable. Paid and credentialed sources are never fetched silently.

Free official and open sources

Source	Category	Local state	Role / route
AVR042: AVR Hardware Design Considerations	mcu-foundation	identity-pinned-private-cache	generic vendor guidance; official HTTPS intake
AN2519: AVR Microcontroller Hardware Design Considerations	mcu-foundation	identity-pinned-private-cache	generic vendor guidance; official HTTPS intake
AN-1142: Techniques for High Speed ADC PCB Layout	mixed-signal-pdn	network-blocked-monitor-online	generic vendor guidance; retry then browser-capable official fallback
AN-136: PCB Layout Considerations for Non-Isolated Switching Power Supplies	switching-power	network-blocked-monitor-online	topology guidance; retry then browser-capable official fallback
Hardware Design Guide for KeyStone I Devices	complex-soc	identity-pinned-private-cache	device-family guidance; official HTTPS intake
MT-101: Decoupling Techniques	pdn-decoupling	network-blocked-monitor-online	generic engineering guidance; retry then browser-capable official fallback
MT-031: Grounding Data Converters and Solving the Mystery of AGND and DGND	mixed-signal-grounding	network-blocked-monitor-online	generic engineering guidance; retry then browser-capable official fallback
High-Speed Layout Guidelines	signal-integrity	identity-pinned-private-cache	generic vendor guidance; official HTTPS intake
High-Speed Interface Layout Guidelines	serdes-high-speed	identity-pinned-private-cache	interface-family guidance; official HTTPS intake
AM62x and AM62Lx DDR Board Design and Layout Guidelines	ddr-memory	identity-pinned-private-cache	device-family guidance; official HTTPS intake
Digital Isolator Design Guide	isolation-emc	identity-pinned-private-cache	generic and family guidance; official HTTPS intake
Best Practices for Board Layout of Motor Drivers	motor-high-current	identity-pinned-private-cache	topology guidance; official HTTPS intake
CC3120 CC3220 SimpleLink Wi-Fi and IoT Solution Layout Guidelines	rf-wireless	identity-pinned-private-cache	device-family reference-layout guidance; official HTTPS intake
Robust Hot Swap Design	power-entry-protection	identity-pinned-private-cache	topology and SOA guidance; official HTTPS intake
Five Steps to a Good PCB Layout of a Boost Converter	switching-power	identity-pinned-private-cache	topology guidance; official HTTPS intake
AN2648: Selecting and Testing 32 kHz Crystal Oscillators for AVR Microcontrollers	clock-oscillator	identity-pinned-private-cache	device-family guidance; official HTTPS intake
AN-0971: Recommendations for Control of Radiated Emissions with isoPower Devices	isolation-emc	network-blocked-monitor-online	device-family guidance; retry then browser-capable official fallback
Ethernet PHY PCB Design Layout Checklist	ethernet-high-speed	identity-pinned-private-cache	interface-family checklist; official HTTPS intake
ESD Protection Layout Guide	esd-protection	already-local-pinned	protection topology guidance; existing local source
UltraScale Architecture PCB Design User Guide	fpga-soc-ddr-serdes	monitored-online-current	device-family authority; version-monitored official portal
Agilex 5 PCB Design Guidelines HSSI EMIF MIPI True Differential PDN	fpga-soc-bga	monitored-online-current	device-family authority; version-monitored official portal
KiCad official 3D model libraries	cad-assets	available-upstream	preferred generic CAD source; GitLab API or pinned clone

Paid standards

Source	Category	Local state	Role / route
IPC-2226A Sectional Design Standard for HDI Printed Boards	hdi-microvia	historical-original-local-current-a-absent	process standard; purchase or user-supplied local intake
IPC-7095D Amendment 1 Design and Assembly Process for BGA	advanced-packaging	not-found-local	package and assembly standard; purchase or user-supplied local intake
IPC-4761 Design Guide for Protection of Printed Board Via Structures	via-protection	not-found-local	process standard; purchase or user-supplied local intake
IPC-6012F Qualification and Performance Specification for Rigid Printed Boards	fabrication-quality	historical-c-local-current-f-absent	qualification and acceptance standard; purchase or user-supplied local intake
IPC-6013E Qualification and Performance Specification for Flexible and Rigid-Flexible Printed Boards	flex-rigid-flex	historical-d-local-current-e-absent	qualification and acceptance standard; purchase or user-supplied local intake
IPC-4552B ENIG Specification	surface-finish	not-found-local	finish specification; purchase or user-supplied local intake

Credentialed and conditional providers

Source	Category	Local state	Role / route
DigiKey Product Information V4	component-metadata	not-configured	discovery and datasheet locator; OAuth API with user credentials
Mouser Search API	component-metadata	not-configured	discovery datasheet lifecycle and supply locator; API key with user account
element14 Product Search API	component-metadata	not-configured	discovery and part metadata locator; API key with user account
Nexar API	component-and-supply-metadata	client-present-no-credentials	discovery and datasheet locator only; OAuth GraphQL with user-approved terms
SnapMagic CAD Platform	cad-assets	not-configured	conditional symbol footprint and 3D source; manual or contracted API with license gate
Ultra Librarian	cad-assets	not-configured	conditional symbol footprint and 3D source; manual account flow or licensed integration